

SYLLABUS

COURSE NAME : MOBILE APPLICATION DEVELOPMENT

COURSE CODE : MCA518-3

COURSE TYPE : Theory

OFFERED PROGRAMMES:

Master of Computer Applications (TRIMESTER -III)

Course learning Outcomes

Outcome Reference No.	Outcome Description	RBT Domain	Highest RBT Level
CO1	Analyze the interactive Android apps with effective UI/UX and navigation.	Cognitive	L4 - Analyse
CO2	Analyze Activity and Fragment lifecycles for performance optimization.	Cognitive	L4 - Analyse
CO3	Evaluate data storage and network communication for efficient handling.	Cognitive	L5 - Evaluate
CO4	Create asynchronous programming for responsiveness and background tasks.	Cognitive	L6 - Create
CO5	Evaluate Android security and best practices for scalable, secure apps.	Cognitive	L5 - Evaluate

Sustainability Development Goals (SDG)

- Industry, innovation and infrastructure

21st Century Skills

- Creative thinking
- Technological literacy

Course Credit Details

Display Name	Course Credit	Hours Per Session
Theory and Practical	4	75

Course Content

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
U1	ANDROID BASICS	15	Global	Skill Development	Professional Ethics	-

Topic No.	Content	Description	Topic Reference Material Upload
U1-T1	History of Mobile Apps, Trends in Market-Web App Vs Mobile App-Mobile OS. Introduction to Android and Kotlin: Kotlin Basics – Classes and Objects- Inheritance- Functions – Extension Functions – First Android App – Anatomy of an Android App - Deploying the app: Running and Debugging app in Android Emulator. Lab Exercises: 1. Setting up Android Studio with emulator / XML Interfaces 2. Activity and Layout demonstration	This unit includes the introduction to Android and Kotlin.	-

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
U2	LAYOUT NAVIGATION	15	Global	Skill Development	Professional Ethics	-

Topic No.	Content	Description	Topic Reference Material Upload
U2-T1	Layouts in Android ConstraintLayout - Displaying lists with RecyclerView Multiple activities and intents - App bar, navigation drawer, and menus Fragments - Navigation in an app - Navigation UI. Lab Exercises: 3. Intents 4. User navigation	This unit explores the Layouts and intents.	-

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
U3	ACTIVITY AND FRAGMENT LIFECYCLE	15	Global	Skill Development	Professional Ethics	-

Topic No.	Content	Description	Topic Reference Material Upload
U3-T1	Introduction to Activity-Activity Lifecycle – Logging. Fragment: Introduction - Lifecycle- Task and Back Stack. Android App Architecture - View Model -Data Binding – Live Data- Transform Live Data. Lab Exercises: 5. Activity Lifecycle 6. Fragment Lifecycle	This chapter incorporates Activity Life Cycle, Fragment and App Architecture.	-

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
U4	SAVING USER DATA	15	Global	Skill Development	Professional Ethics	-

Topic No.	Content	Description	Topic Reference Material Upload
U4-T1	Store Data-Room Persistency Library- Asynchronous program-Coroutines-Testing Databases. Introduction to Advanced Binding – Multiple Item View types-Headers -GridLayouts. Lab Exercises: 7. Sharedpreference 8. Recyclerview	This unit incorporates Data Storage and Multiple Item View	-

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
U5	ADVANCED RECYCLERVIEW	15	Global	Skill Development	Professional Ethics	-

Topic No.	Content	Description	Topic Reference Material Upload
U5-T1	Connect to the Internet-Android Permissions-connect to and from Network Resources – Connect to the Web Services-Display Images. Repository pattern – Work Manager – Work Input/Output – Work Request Constraints. App UI Design: Android Styling – Typography-Material Design- Material Components-Localization. Lab Exercises: 09. Work Manager 10. Material Design	This unit contains Android Permissions, Web Services, App UI Design.	-

Text & Reference Books

[1] John Horton, Android programming with Kotlin for beginners, Packt-Birmingham, Mumbai, 2nd edition, 2019.

[2] Gardner, B., Sills, B., Stewart, C., Marsicano, K. Android Programming: The Big Nerd Ranch Guide. United Kingdom: Addison Wesley Professional, 4th edition, 2022

Essential Reading

1. Dawn Griffiths and David Griffiths, Head First Android Development: A Brain-Friendly guide, O'Reilly, 2nd edition, 2019. Mark Wickham, Practical Android: 14 Complete Projects on Advanced Techniques and Approaches, APRESS.

EXAMINATION PATTERN

PATTERN FOR CONTINUOUS INTERNAL ASSESSMENT (CIA)

CIA includes Lab Assessments, Regular Lab Exercise and theory Test

QUESTION PAPER PATTERN FOR END SEMESTER EXAMINATION (ESE)

Either or pattern